

Linjun Shou

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SUMMARY

Partner Group Applied Scientist Manager currently leading the AI efforts of **Edge browser** and **Office Agent (project codename "Societas")**, a team of 30+ engineers and scientists. Extensive experience in search technologies — ranking, document understanding, query understanding (QU), and Q&A systems. Published dozens of papers in top-tier research conferences and holds 20+ U.S. patents. **Two-time recipient of the Distinguished Contribution Award** from the CTO office at Microsoft MLADS (Spring and Fall 2021).

Expertise: Machine Learning, Copilot, Edge Intelligence, Document Understanding, Search, Question Answering, Recommendation, Natural Language Processing.

EDUCATION & EMPLOYMENT

Edge ML and Societas ML Team, Microsoft Redmond <i>Partner Group Applied Science Manager</i>	2025 – Present
Edge Machine Learning, Microsoft China <i>Principal Group Applied Scientist Manager</i>	2022 – 2025
Bing Search, Microsoft STCA <i>Principal Applied Scientist Manager</i>	2015 – 2022
Institute of Computing Technology, Chinese Academy of Sciences <i>Master's degree, Computer Sciences (Honored Graduate)</i>	2012 – 2015
Shanghai University <i>Bachelor's degree, Mathematics (Top 1 / 297 in 4 consecutive years)</i>	2008 – 2012

WORK & PROJECTS

Office Agent — Microsoft 365 Copilot (project codename "Societas") <i>Societas ML Team, Microsoft Redmond</i>	2025 – Present
• Motivation & Goal: Build a multi-agent system for Microsoft 365 Copilot that delivers polished PowerPoint presentations, ready-to-use Word documents, and (soon) Excel spreadsheets — handling complex knowledge-work workflows end-to-end. • Architecture: Multi-agent orchestration engine on an open-source agent stack with Anthropic's Claude — a central planner agent coordinates tasks and synthesizes results; specialized agents for code, finance, search, etc. work in parallel; a secure tool layer integrates utilities and sandboxed environments. Adopted a code-first approach, letting the model write and execute code (e.g., MP3 transcription, PDF text extraction) instead of relying on task-specific tools, for greater flexibility and generalization. • Taste-Driven Development (TDD): pioneered a new generation paradigm — distilled "taste blueprints" from high-quality, in-house presentation/document samples; injected them into planning and execution to drive consistent design language. Built auto-theming (theme generated from content rather than picked from templates) and a content self-verification + iterative refinement loop. Output is HTML5-based slides with a converter to native PowerPoint. • Benchmark — GAIA & TDDEval (SOTA): achieved state-of-the-art on the public GAIA benchmark — L1 88.7 / L2 76.7 / L3 60.0, ahead of Genspark (87.8 / 72.7 / 58.8), Manus (86.5 / 70.1 / 57.7), and OpenAI Deep Research (74.3 / 69.1 / 42.3) per officially reported metrics. Also defined the in-house TDDEval benchmark for taste-driven generation across PowerPoint, Excel, and Word, scoring on Content Quality (grounding, relevance, completeness, structure, usability) and Taste Score (visual appeal, layout, typography, design consistency, asset curation). • AI-native execution: built and shipped the system as a fully AI-native team of 7, going from zero to SOTA in just 8 weeks, with 95%+ of code generated and reviewed via AI coding tools.	

- **Deliveries:** SOTA on GAIA across L1/L2/L3 difficulty tiers; Office Agent shipped as a multi-agent capability for Microsoft 365 Copilot. [Microsoft Tech Community blog](#).

Copilot in Edge Browser

2023/03 – 2024/05

Edge ML, Microsoft

- **Motivation & Goal:** Lead the team to build the most intelligent browser with Edge Copilot.
- **Accountability:** Fully responsible for the Edge Copilot algorithm and feature development — (1) Contextual chat with any documents in browser; (2) Video Copilot — chat with any videos; (3) Edge Actions for end-to-end task completion; (4) Personal Copilot — ask anything about what you have browsed before.
- **Deliveries:** (1) Chat DAU reached **3 million (10× increase)**; (2) Contextual chat supports 1000+ pages long documents at **95%+ accuracy**; (3) Quoted by Microsoft CEO Satya Nadella: "*my most favorite agentic experience is Tab auto-grouping in Edge Copilot*"; (4) Strong positive user feedback (e.g., "*God bless those who invented Edge Copilot*" on Twitter).

Copilot in Windows

2023/03 – 2024/05

Edge ML, Microsoft

- **Motivation & Goal:** Ensure agile shipping of the first Windows Copilot release; Edge team joined forces with Windows team. Windows Copilot was built on top of the Edge Copilot codebase.
- **Responsibility:** My team owned the back-end and algorithm development to ship Copilot in the world's largest operating system.
- **Deliveries:** (1) Shipped the first Windows Copilot release with positive user feedback; (2) Shipped Windows Skills powered by the Edge Action framework I built; (3) Shipped contextual chat in Windows Copilot; (4) Shipped the screen understanding feature — see what you see on your screen via screenshot.

Edge Client Intelligence — Client DNN Models

2022/06 – 2023/03

Edge ML, Microsoft

- Developed extreme model compression technologies (distillation, quantization, etc.) to ship multiple client DNN models — **DeepEE, DeepPU, DeepAutoFill, DeepUU** — powering AI features in Edge with enhanced privacy and COGS savings.
- **DeepEE:** shipped the very first client universal transformer models in Edge with P/R over 95, to **69 languages, 154+ regions worldwide** for PH and PC, powering Shopping globalization. **4M peak DAU** in shipping season.
- **DeepPU:** shipped to 6+ scenarios — shoreline personalization, 1CNS personalized notifications, contextual prompt suggestions, shopping PDP triggering, tab group auto-naming and auto-grouping, etc.
- **DeepAutoFill:** autofilling scenarios in Edge.
- **DeepUU:** shipped V1 to build client user profile in global markets, powering Edge personalization with **BSoM proxy gains +0.79% (abs) and +2% (rel)**.

Document Understanding for 400B Web Index

2019 – 2022

Document Understanding Team, WebXT, Microsoft

- Led the STCA DU team (16 engineers/scientists) to develop the best document understanding DNN techniques and platform for search and recommendation scenarios.
- Developed **Universal Multi-modal Information Extraction** (Semantic Document) powering 15+ production scenarios — Caption, QnA, Ads, Recommendation. Extended to fine-grained extraction (product name, price, manufacturer, image).
- Developed **Universal Topic Understanding** models with 100k topics powering 15+ scenarios — Sage News, Windows Feeds (50+ regions), Underside, Enterprise News, Bing QnA, PAA.
- Developed SOTA document full-body representation model (FBV) — shipped to Bing search for **400 billion vector space ANN search**.
- Developed Long Document MRC: one of the longest DU models in 100+ languages, powering 8+ scenarios — Azure, QnA, PAA, MSN News, Knowledge Card.
- Developed authority techniques (topical authority and host embedding) powering Bing QnA, PAA, core relevance, spam detection, personalized search.
- Excellent team collaboration — received 16 positive partner feedbacks (VP, Partner, Principal level) in FY21 across multiple orgs.

Recommendation Algorithms for Feeds

2020

ASK & DU Team, Bing, Microsoft

- **QnA Feeds:** as tech lead, drove the recommendation streaming index gen and ranking system (3 layers of ranking and two-tower embedding model) from scratch.
- **News Recommendation:** developed and shipped topic/KPE-based personalized recall and topic-based diversity for the first Windows Feeds release in 50+ regions on Windows 10/11.
- **Web Recommendation:** as tech lead, drove FBV techniques for semantic recall path — **+21 points recall@100**.
- **Movie Recommendation:** developed knowledge-graph-enhanced collaborative filtering — **+10 points**.

Bing QnA Universalization in Global Regions

2019 – 2020

ASK Team, Bing, Microsoft

- As tech lead, scaled QnA from English-only to **100+ languages and 200+ regions** for the very first time.
- Proposed a systematic **language scaling methodology** for low-resource languages — **2× MLADS Distinguished Contribution Award (top 2 / 300, record-breaking)**.
- Designed and drove a brand-new universal streaming QnA pipeline from scratch — improved QnA refresh cycle from **1 month to minutes/seconds** and saved **80% of GPU capacity**.

Deep Learning Technologies for Bing QnA

2017 – 2019

Quantus Team, Bing, Microsoft

- Shipped 10+ DNN releases improving core QnA relevance — beat and replaced legacy GBDT ranker (400+ hand-crafted features) for the first time and won WPSBS vs. Google.
- Designed and built DNN-powered full-stack list QnA ranking pipeline; shipped 10+ releases contributing to **+20 precision points** to beat Google.
- Proposed novel model compression techniques for heavy pre-trained model online serving [Shou2020RSKD, 15], powering the **first BERT model released in Bing** (within 1 month of BERT open-source) — +3.0% coverage and +5.0% precision.
- Open-sourced an NLP DNN toolkit, **NeuronBlocks**, on GitHub — **1k+ stars in 2 months**; reduced model dev time from weekly/daily to minutes.

Bing News and Tweet Core Ranking

2015 – 2016

News Team, Bing, Microsoft

- Shipped dozens of ranker releases in Bing News ranking and News homepage ranking.
- Shipped ML-based Tweets ranker — improved Twitter answer defect rate from **18% to 2%**.
- Designed breaking-news detection system — minute-level latency (from daily); filed a US patent.
- Developed and shipped Rich Nav news feature on Bing from scratch — 2.73 strong surplus, 19.17 weak surplus.

SELECTED PUBLICATIONS (50+ papers, 9,500+ citations on Google Scholar — full list on [scholar profile](#))

1. Nuo Chen, **Linjun Shou**, Tengtao Song, Ming Gong, Jian Pei, Jianhui Chang, Daxin Jiang, Jia Li. "Structural Contrastive Pretraining for Cross-Lingual Comprehension." *ACL* 2023. (intern mentored)
2. Nuo Chen, **Linjun Shou**, Ming Gong, Jian Pei, Bowen Cao, Jianhui Chang, Daxin Jiang, Jia Li. "Alleviating Over-smoothing for Unsupervised Sentence Representation." *ACL* 2023. (intern mentored)
3. Shengyao Zhuang, **Linjun Shou**, Jian Pei, Ming Gong, Houxing Ren, G. Zuccon, Daxin Jiang. "Typos-aware Bottlenecked Pre-Training for Robust Dense Retrieval." *SIGIR* 2023. (intern mentored)
4. Nuo Chen, **Linjun Shou**, Ming Gong, Jian Pei, Daxin Jiang. "Bridging the Gap between Language Models and Cross-Lingual Sequence Labeling." *NAACL* 2022. (intern mentored)
5. **Linjun Shou**, Ming Gong, Jian Pei, Xiubo Geng, Xingjie Zhou, Daxin Jiang. "Language Scaling: Applications, Challenges and Approaches." *KDD Tutorial* 2021.
6. **Linjun Shou**, Ming Gong, J. Pei, Xiubo Geng, Xingjie Zhou, Daxin Jiang. "Scaling out NLP Applications to 100+ Languages." *WWW Tutorial* 2021.
7. S. Liang, M. Gong, J. Pei, **Linjun Shou**, D. Jiang. "Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition." *KDD* 2021. (intern mentored)
8. Yingmei Guo, **Linjun Shou**, Jian Pei, Ming Gong, Mingxing Xu, Zhiyong Wu, Daxin Jiang. "Learning from Multiple Noisy Augmented Data Sets for Better Cross-Lingual Spoken Language Understanding." *EMNLP* 2021.
9. Junjie Huang, Duyu Tang, **Linjun Shou**, Ming Gong, Ke Xu, Daxin Jiang, Ming Zhou. "CoSQA: 20,000+ Web Queries for Code Search and Question Answering." *ACL* 2021.

10. **Linjun Shou**, S. Bo, Fei-Xiang Cheng, Ming Gong, J. Pei, Daxin Jiang. "Mining Implicit Relevance Feedback from User Behavior for Web Question Answering." *KDD* 2020.
11. Shining Liang, **Linjun Shou**, Jian Pei, Ming Gong, Wanli Zuo, Daxin Jiang. "CalibreNet: Calibration Networks for Multilingual Sequence Labeling." *WSDM* 2021. (intern mentored)
12. Junwei Liao, Yu Shi, Ming Gong, **Linjun Shou**, Sefik Eskimez, L. Lu, H. Qu, Michael Zeng. "Generating Human Readable Transcript for Automatic Speech Recognition with Pre-trained Language Model." *ICASSP* 2021.
13. Junhao Liu*, **Linjun Shou***, Jian Pei, Ming Gong, Min Yang, Daxin Jiang. "Cross-lingual Machine Reading Comprehension with Language Branch Knowledge Distillation." *COLING* 2020. (* equal contribution)
14. F. Yuan, **Linjun Shou**, X. Bai, M. Gong, Y. Liang, N. Duan, Y. Fu, Daxin Jiang. "Enhancing Answer Boundary Detection for Multilingual Machine Reading Comprehension." *ACL* 2020. (intern mentored)
15. Ze Yang*, **Linjun Shou***, Ming Gong, Wutao Lin, Daxin Jiang. "Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System." *WSDM* 2020. (* equal contribution)
16. H. Huang, Yaobo Liang, N. Duan, Ming Gong, **Linjun Shou**, D. Jiang, M. Zhou. "Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks." *EMNLP/IJCNLP* 2019.
17. Ming Gong, **Linjun Shou**, Wutao Lin, Zhijie Sang, Quanjia Yan, Ze Yang, D. Jiang. "NeuronBlocks — Building Your NLP DNN Models Like Playing Lego." *EMNLP/IJCNLP* 2019.
18. Xuguang Wang, **Linjun Shou**, Ming Gong, Nan Duan, Daxin Jiang. "No Answer is Better Than Wrong Answer: A Reflection Model for Document Level Machine Reading Comprehension." *EMNLP* 2020.

PATENTS

- Personalized Multilingual Content Recommendation based on Robust Feature Network. 2023. (MS#412659-CN-NP)
- Split-Screen Viewing Based On Intelligent Triggering. 2023. (MS#413279-CN-NP)
- Sequential Recommendation based on Cross-Domain Behavior Data. 2023. (MS#412874-CN-NP)
- Universal Deep Entity Extraction (DeepEE) for Browser. 2023.
- Providing An Activeness Indication Of A Tab. 2023. (MS#412658-CN-NP)
- Establishing a Language Model Adapted to a Cross-Lingual Sequence Labeling Task. 2022. (MS#411268-CN-NP)
- Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022. (MS#411267-CN-NP)
- Sequential Recommendation Based On Dual Contrastive Interest Learning. 2022. (MS#411269-CN-NP)
- Sentence Representation Generation for Cross-lingual Retrieval. 2022. (MS#411088-CN-NP)
- Content Recommendation Based On Graph Enhanced Collaborative Filtering. 2022. (MS#410901-CN-NP)
- Controversial Poll Auto Generation by Generation Model and Crowdsourcing. 2021.
- Knowledge-enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension. 2021.
- Graph based Fact Clustering for Diversity and Freshness. 2021.
- Representation Learning of Cross-Language Texts. 2021. (MS#409565-CN-NP)
- Hierarchical Representation Learning of User Interest. 2021. (MS#410462-CN-NP)
- Question Generation from Queries. 2021. (MS#410241-CN-NP)
- Multi-Model Joint Denoising Training. 2021. (MS#409566-CN-NP)
- Representation Learning of Semi-structured Data. 2020. (MS#408908-CN-NP)
- Model Selection Learning for Knowledge Distillation. 2020. (MS#408426-CN-NP)
- Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020. (MS#407927-CN-NP)
- Contrastive Learning for Question Answering. 2020. (MS#407988-CN-NP)
- Model Generation based Model Compression. 2019. (MS#406805-CN-NP)
- Cross-Lingual Task Training. 2019. (MS#406497-CN-NP)
- Assertion-based Question Answering with Question-Aware Open Information Extraction. 2019.
- EventFactory: detect event's scale, sub-intent, category, timeline, social sentiment with AI for search and recommendation. 2016. (US WO2018098751A1)

AWARDS

- Microsoft 2024 Hackathon Greater China Region — **Outstanding Project Award**: Bootstrap Copilot Pages with Edge Browsing Task Center. (Won 4 awards in total — record-breaking.)
- Microsoft 2024 Hackathon Beijing — Outstanding Project Award: Browse for Me.
- Microsoft 2024 Hackathon Beijing — Outstanding Project Award: Microsoft Mindmap — Transform Anything into Mind Maps.

- Microsoft 2024 Hackathon Suzhou — Outstanding Project Award: Tango — On-Device AI Service Stack for Windows M365 Apps.
- **Microsoft 2021 MLADS Fall Distinguished Contribution Award (2 / 250+)**. First time in history to receive the Award twice in 2 consecutive conferences.
- **Microsoft 2021 MLADS Spring Distinguished Contribution Award (2 / 300+)**.
- Excellent Service as Teaching Assistant of AI School China — 2019, 2020, 2021 (won Excellent Project Award — 1st Prize twice).
- Microsoft AI Core Q3 FY18 Greatness Award. 2018.
- Excellent Student of CAS (Top 5%). 2013.
- Honored Graduate of Shanghai (Top 1%). 2012.
- National Scholarship (Top 1%). 2011.
- First Prize Scholarship in SHU (Top 1%) — 4 consecutive times. 2008–2012.